

Christopher A. Jones

cjones2673@gmail.com | <https://cjones.dev> | Morgantown, WV

PROFILE

Computer Science and Cybersecurity student graduating December 2026, passionate about both technology and education. I build full-stack and distributed systems and run my own Linux servers, and between teaching kids robotics and tutoring college CS students, building software that helps people learn is the work I most want to do.

EDUCATION

West Virginia University, Morgantown, WV · Honors College · Expected December 2026

B.S. Computer Science and B.S. Cybersecurity · GPA 3.79 · President's List (Fall 2025)

Relevant coursework: Operating System Structure, Web Application Design, Analysis of Algorithms, File and Data Structures, Secure Software Development

TECHNICAL SKILLS

- **Languages:** Java (strongest), JavaScript (ES6), SQL, C, Python, Go, HTML/CSS
- **Front-end:** React, Vite, Node.js
- **Back-end & data:** .NET 8, Apache Spark, Apache Hadoop, MySQL, Google Firestore, OAuth 2.0
- **Tools & platforms:** Linux/Unix (Debian, Ubuntu, Fedora, Arch), Apache HTTP Server, Git/GitHub, Docker / Docker Swarm, Wireshark, QEMU

PROJECTS

PICARD: Distributed Machine-Learning Platform · Senior capstone, 5-person team · Spring 2026–present

React · .NET 8 · Apache Spark · Apache Hadoop · Docker Swarm · MySQL · OAuth 2.0

- Lets researchers run ML experiments on a Hadoop/Spark cluster without managing the infrastructure: upload an algorithm, dataset, and parameters, and the platform handles authentication, queueing, distributed execution, and results.
- Accepts researcher-submitted algorithms as Scala JARs or Python scripts and runs them across the cluster.
- Designed the Parameter Sweep Engine: server-side batch generation that expands a range of parameters into one atomic, trackable batch.
- Produced architecture diagrams and mapped critical code paths across the React, .NET, and Spark stack to guide implementation.

TaskGarden: Gamified Task Manager · CS 230 team project (5 people) · Spring 2024

React · Firestore · Firebase Hosting

- Built a task-management web app where completing tasks grows a virtual plant; implemented the React front end on a Firestore data model and shipped a hosted product over a semester with teammates.

portmango: Network Mapper in Go · Personal · 2026–present

- Re-implementing core discovery techniques (ARP host discovery, TCP SYN/connect scanning, ICMP probing) from scratch in Go with JSON-first output, verified in Wireshark against a containerized lab network.

MergeConflict (MPX): Operating System from Scratch · CS 450 team project · Fall 2024

- Built a small multiprogramming OS in C and x86 assembly (process scheduling, memory allocation, interactive command handler, device I/O) without the C standard library, tested under QEMU.

Self-Hosted Homelab · Personal · ongoing

- Administer two Debian servers (a persistent TCP game server and a development sandbox) with self-hosted Apache pages; comfortable with Linux/Unix administration and networking.

EXPERIENCE

Tutor, LCSEE Learning & Mentoring Center · West Virginia University · Sept 2024–Present

- Provide free, one-on-one help to Computer Science, Cybersecurity, and Computer Engineering students across programming, algorithms, and systems courses, coaching debugging and problem-solving strategies that build independent learners.

Robotics Instructor Intern · NASA Katherine Johnson IV&V Education Resource Center, Fairmont, WV · Summers 2023 & 2024

- Taught robotics programming to elementary and middle-school students (ages 6–12) on the VEX IQ platform, designed engaging STEM content, and helped organize a regional drone competition; recognized for adaptability, reliability, and communication.

RESEARCH

Generative AI Use for Programming Work · Research report (unpublished) synthesizing scholarly research and faculty interviews on generative AI's impact on programming, weighing productivity gains for experienced developers against over-reliance risks for novices.